

```
def ends_with_ab(string):  
    state = 0  
  
    for char in string:  
        if state == 0:  
            if char == 'a':  
                state = 1  
            else:  
                state = 0  
  
        elif state == 1:  
            if char == 'b':  
                state = 2  
            elif char == 'a':  
                state = 1  
            else:  
                state = 0  
  
        elif state == 2:  
            if char == 'a':  
                state = 1  
            else:  
                state = 0  
  
    return state == 2  
  
# Test Cases  
strings = ["ab", "aab", "cab", "abab", "abc", "baa"]  
  
for s in strings:  
    if ends_with_ab(s):  
        print(f"{s} -> Accepted")  
    else:  
        print(f"{s} -> Rejected")  
|
```

 IDLE Shell 3.14.3

File Edit Shell Debug Options Window Help

Python 3.14.3 (tags/v3.14.3:323c59a, Feb 3 2026, 16:04:56) [MSC v.1944 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.

>>>

===== RESTART: C:/Users/sreel/OneDrive/Desktop/experiment 1-

ab -> Accepted

aab -> Accepted

cab -> Accepted

abab -> Accepted

abc -> Rejected

baa -> Rejected

>>>